

Emily Zhao

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Education

STANFORD UNIVERSITY

B.S. in Electrical Engineering, Minor in Energy Sciences & Engineering

Class of 2028

GPA: 3.75

Relevant Coursework: Linear Algebra, Special Relativity, Differential Equations, Quantum Mechanics, Statistical Mechanics, Data Structures, GIS Planning for Renewables & Transmission, Circuits, Signals Processing, Probability

Experience

AEROSPACE CONTROLS ENGINEER | *Laminar Vector*

Shenzhen, China | 07/26 – 09/26

- Developed C++/PX4 flight-control systems for a novel large-scale airframe using Pixhawk FMUs and QGroundControl; evaluated aircraft dynamics and closed-loop control behavior through Simulink and software-in-the-loop workflows.
- Supported airframe prototyping, manufacturing, and flight-test operations at Shenzhen factories.

DATA SCIENTIST | *Halcyon*

San Francisco, CA | 03/26 – 08/26

- Developed and conducted energy-market research for LLM-powered data products to structure utility, grid, and regulatory data into interactive interfaces spanning IRPs, rate cases, generation projects, transmission infrastructure, and energy markets.
- Engineered and evaluated RAG/data pipelines across PostgreSQL, REST APIs, asynchronous batch processing, document retrieval, and structured LLM outputs; benchmarked model responses against ground-truth datasets.

GEOPHYSICS RESEARCH FELLOW | *Bruno Kessler Foundation*

Trento, Italy | 06/24 – 08/24

- Developed an ML-based geophysical anomaly detection pipeline and web prototype for investigating and visualizing potential earthquake precursors using satellite and multi-source environmental data.
- Built and integrated a multimodal dataset spanning CSES-01 satellite, ionospheric, solar, cosmic-ray, and ground-based geophysical data; developed a multi-head Transformer model to identify spatiotemporal anomalies preceding seismic events.

Projects

Ultrathin Semimetal Interconnects Research | Eric Pop Lab, Stanford University

09/26 – Present

- Fabricate ultrathin NbP/Nb heterostructures and microdevices for next-generation nanoelectronic interconnects, using thin-film deposition and cleanroom fabrication techniques to prepare samples for electrical testing.
- Characterize structural and electrical properties of nanoscale NbP films, investigating thickness-dependent resistivity, carrier transport, and surface-dominated conduction to evaluate performance beyond conventional metal interconnects.

Magnetic Nanomaterials Research | Shan X. Wang Lab, Stanford University

6/26 – 09/26

- Research FeCo magnetic nanorods for >28 GHz RF devices, engineering high saturation magnetization and shape anisotropy.
- Synthesize via solution-phase polyol chemistry, investigating precursor chemistry, temperature, diffusion, and reaction kinetics.
- Support fabrication and characterization of magnetically aligned nanorod arrays, relating nanoscale morphology and magnetic anisotropy to high-frequency properties through techniques including TEM/SEM, VSM, and FMR measurements.

Custom Quadcopter Flight Controller & Wireless Control System

06/26 – 9/26

- Built a custom ESP32/C++ quadcopter flight controller and Fusion 360 airframe with PID stabilization, IMU feedback, 4-in-1 ESC motor control, SPI/UART peripherals, power distribution, and a custom wireless transmitter–receiver system.
- Integrated propulsion, power, sensing, wireless communication, embedded controls in a complete electromechanical system.

Piezocatalytic Hydro–Turbine: Materials, CAD & CFD Optimization

09/24 – 5/24

- Designed crossflow hydro-turbine in CAD, using CFD simulations to optimize bluff-body geometry, vortex shedding, pressure fields, and coupled materials–fluid system interactions for enhanced piezoelectric deformation and energy transfer.
- Synthesized and characterized MoS₂/PVDF piezocatalytic films; experimentally validated designs across flow conditions.

Skills

Programming & Data: Python, C++, C, SQL, MATLAB, Git, PostgreSQL, REST APIs, Pandas, GIS

Embedded Systems & Controls: Arduino, PX4 Autopilot, Pixhawk/FMUs, PID Control, IMU, SPI, UART, CAN, QGroundControl

Engineering & Simulation: Autodesk Fusion 360, Simulink, CFD, CAD, 3D Printing, Soldering & PCB Prototyping, Oscilloscopes, Multimeters, AC/DC Power Supplies, Signal/Function Generators, Spectrum Analysis

Materials & Characterization: Nanoparticle Synthesis, Magnetic Nanomaterials, SEM/TEM, VSM, Ferromagnetic Resonance, Thin-Film Fabrication, Sputtering, Photolithography, Electrical Transport, Hall Measurements, SEM, AFM, XPS/EDS

AI & Data Systems: LLM/RAG Evaluation, Structured LLM Outputs, Retrieval Pipelines, Pydantic, RAGAS